

Interim Report Preview

1 02_CNN_Introduction

1.1 Task 1: Classification

Report (classification):

- **Bar figure and discussion.** We compared four CNN designs on CIFAR-10 (Figure ??). Depth and max pooling gave the largest gains; average pooling underperformed. Training accuracy is consistently higher than validation accuracy, indicating mild overfitting as model capacity increases.
- **Best architecture sketch.** The best design was 3conv_32_64_128_max (Figure ??).

figs/cnn_task1_bar.png

Figure 1: Training vs. validation accuracy for four CNN design choices (Task 1).

figs/cnn_task1_arch.png

Figure 2: Sketch of the best architecture: 3conv_32_64_128_max.

Table 1: Best training/validation accuracy (%) per design choice.

Design	Train	Val
3conv_32_64_128_max	53.15	52.72
3conv_32_64_128_leaky	52.35	52.36
3conv_32_64_128_avg	44.96	46.35
2conv_32_64_max	39.65	39.51

1.2 Task 2: Regression

Report (regression):

- **Architecture with validation error < 75%.** All three CNN regressors achieved validation MAPE below 75%, and the widest model performed best (Table ??).
- **Training/validation loss vs. epochs.** Loss curves are shown in Figure ?? . The wider model reduces validation error but increases the train-val gap later in training.
- **Other house sections.** Results for kitchen, bedroom, and bathroom are reported in Table ??.

figs/cnn_task2_loss.png

Figure 3: Training and validation loss vs. epochs for regression architectures.

Table 2: Best validation MAPE (%) for regression models and room sections.

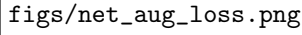
Architecture	MAPE	Section	MAPE
wider_32_64_128	43.59	frontal	49.64
medium_16_32_64	46.17	kitchen	50.23
small_16_32	48.51	bathroom	57.25
		bedroom	59.08

2 03_Network_Training

2.1 Task 1: Tuning a Classification Model

Report (tuning a classification model):

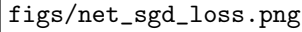
- **Data augmentation.** We trained the baseline without augmentation and two strategies: **Aug1** = RandomCrop(32, padding=4) + RandomHorizontalFlip; **Aug2** = RandomAffine(15°, translate 0.1, scale 0.9–1.1) + RandomHorizontalFlip + ColorJitter (brightness/contrast/saturation 0.2). Loss curves are in Figure ?? . Aug1 achieved the best validation accuracy (81.71%), Aug2 was slightly lower but both improved over no augmentation (Table ??).



figs/net_aug_loss.png

Figure 4: Training/validation loss for no augmentation, Aug1, and Aug2.

- **Dropout and BatchNorm.** Without augmentation, we added Dropout2d after each conv block and before the final linear layer with $p = 0.3$, and BatchNorm after each convolution. These regularizers reduced validation accuracy relative to the baseline (Table ??), suggesting underfitting at the chosen settings.
- **Zero initialization.** Using zeros for kernel initialization produced 10% validation accuracy (random guess on CIFAR-10; Table ??).
- **SGD learning rate sweep.** Using SGD with learning rates $\{3e-3, 1e-3, 3e-4\}$, the loss curves are shown in Figure ??; higher learning rates converge faster early but show more oscillation.



figs/net_sgd_loss.png

Figure 5: Training/validation loss for SGD learning rates $3e-3$, $1e-3$, and $3e-4$.

Table 3: Best validation accuracy (%) for augmentation and regularization settings.

Setting	Best Val Acc
aug1	81.71
aug2	81.42
no_aug	79.37
baseline+batchnorm+dropout	76.83
baseline+dropout	76.79
baseline+batchnorm	76.02
zeros_init	10.00